KOÇ UNIVERSITY
COLLEGE OF ENGINEERING

MECH 303

Machine Design

Project 1:

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Objective

The primary objective of this project was to investigate and quantify the stress concentration factor (Kt) in a flat plate under two distinct loading conditions using the Finite Element Method (FEM).

The two scenarios analyzed were:

1. **Task 1 (Unloaded Hole):** A plate with a central transverse hole subjected to uniform uniaxial tension applied at its ends.
2. **Task 2 (Pin-Loaded Hole):** A plate (bracket) loaded via a pin through its central hole, simulating a clevis or lug joint.

The goal was to build a parametric FEA model in ANSYS to compute the maximum stress (sigma_max) for a range of hole diameters (d). This data was then processed in MATLAB to plot the relationship between the stress concentration factor (Kt) and the geometric ratio (d/b), and to compare these two loading scenarios against established theoretical literature.

Methodology

The analysis was conducted using ANSYS Workbench 2025, utilizing SpaceClaim for geometry, ANSYS Mechanical for simulation, and MATLAB for data processing and visualization.

2.1. Task 1: Unloaded Hole Analysis

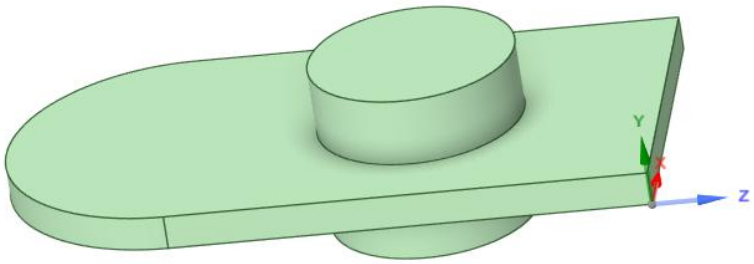
- **Geometry and Parameterization:** Initial geometry modeling was attempted in SolidWorks. However, challenges were encountered in reliably passing the design parameters to the ANSYS Workbench environment. To resolve this, the model was recreated natively in ANSYS SpaceClaim. A flat plate with dimensions b = 50 mm, h = 5 mm, and L = 140 mm was modeled. The central hole diameter, d, was designated as a 'driving dimension', making it an input parameter for the simulation.
- **FEA Setup (ANSYS Mechanical):**
    1. **Material:** Standard 'Structural Steel' was assigned to the plate.
    2. **Meshing:** A default mesh was generated and then refined. A local 'Sizing' control was applied to the circular edges of the hole to ensure a high element density in the region of the expected high-stress gradient.

3. **Boundary Conditions (BCs):** To simulate uniform uniaxial tension, a 'Fixed Support' was applied to one end-face of the plate. A 'Force' of 12,000 N was applied to the opposite end-face in the axial direction (Z-axis).
4. **Parameterization:** The maximum 'Normal Stress' on the plate was defined as an output parameter. The ANSYS Parameter Set was configured to solve for 8 distinct d values, ranging from d=3.75 mm to d=30 mm.

2.2. Task 2: Pin-Loaded Hole Analysis (Theoretical Comparison)

The methodology for the second task involved modeling a more complex, non-linear system.

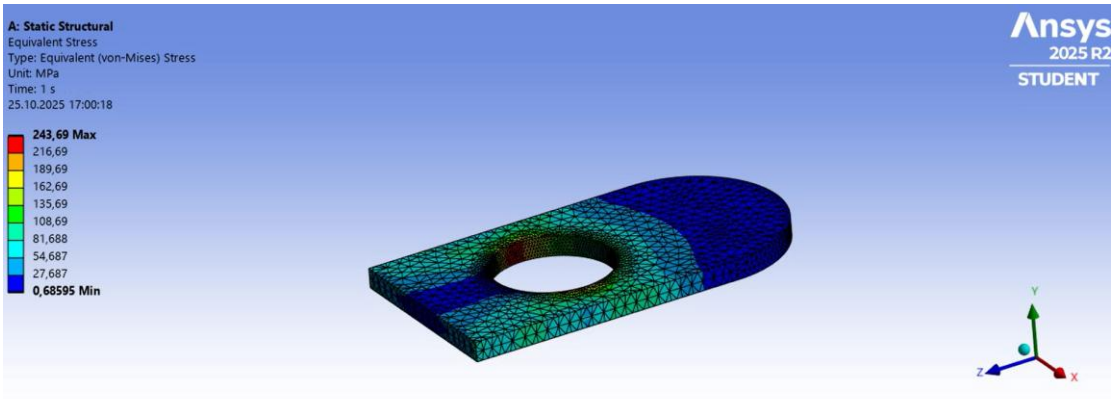
- **Geometry and Material:** An assembly of two bodies, the Bracket and the Pin, was created in SpaceClaim. The pin diameter and the hole diameter were both linked to the same parameter d to create a line-to-line fit. A new material, Structural Steel 2 (with E = 2 x 10^7 MPa), was created and assigned to the pin by duplicating the Structural Steel.



- **FEA Setup (ANSYS Mechanical):**
    1. **Contact:** A 'Frictional' contact pair (with mu = 0.2) was defined between the inner face of the hole (Target) and the outer face of the pin (Contact). The 'Adjust to Touch' option was enabled to ensure contact was established.
    2. **Meshing:** Local sizing was applied to the two contacting faces to improve contact detection and accuracy. Problem was Student

edition of Ansys didn’t allow for many nods and elements so I strategically placed the nods especially around the hole.

3. **Boundary Conditions (BCs):** The 'Fixed Support' was applied to the end-face of the bracket. Two 6,000 N force was applied axially (Z-direction) to the end-faces of the pin.
4. **Stabilization:** This non-linear model was prone to rigid body motion. To stabilize the solution, a 'Displacement' constraint was applied to the pin's end-faces, setting X and Y translations and all three rotations to zero, leaving only the Z-translation (the direction of force) free.
5. **Solution Settings:** 'Large Deflection' was set to 'On', and 'Auto Time Stepping' was enabled to allow the solver to apply the load incrementally and converge.



2.3. Post-Processing and Plotting

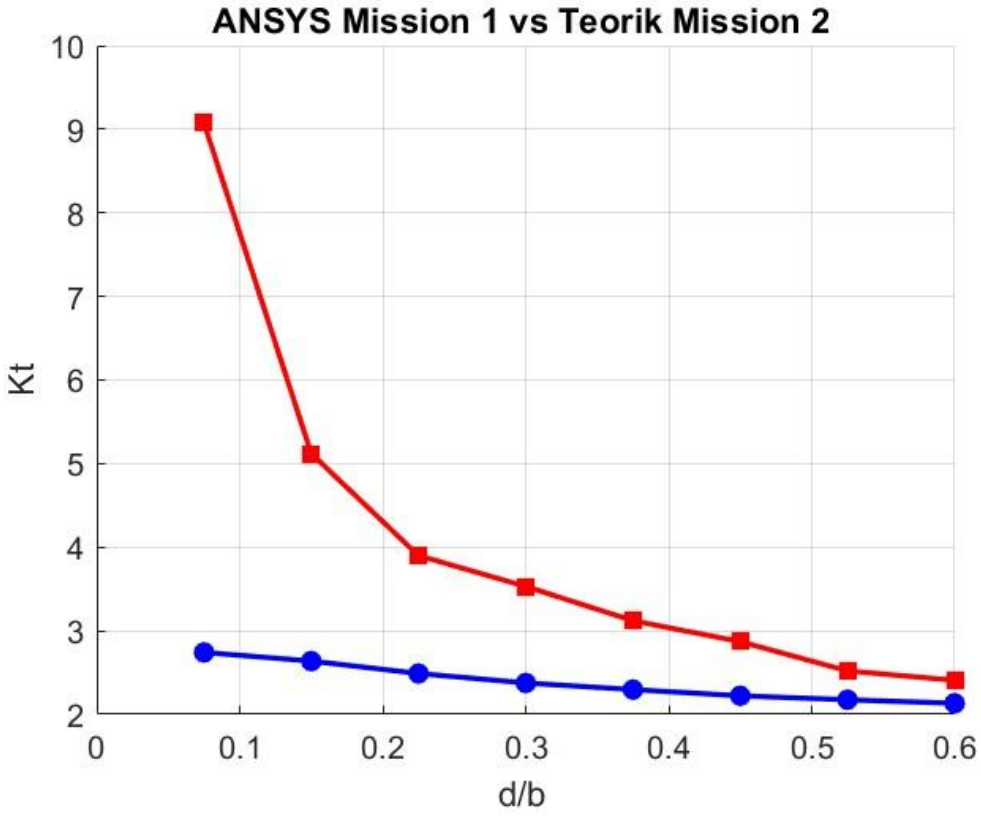
- **ANSYS Data:** The 8 sigma_max values for each task from the ANSYS simulation were exported to MATLAB.
- **MATLAB Calculation:** For both datasets, the d/b ratio and the nominal stress (sigma_nom = F / ((b-d)h)) were calculated. Kt was then determined using the formula Kt = sigma_max / sigma_nom.
- **Plotting:** The Kt vs. d/b curves for both 'Task 1 (ANSYS)' and 'Task 2 (Theoretical)' were plotted on a single figure for direct comparison.

Results

The primary output of this project is the comparative graph generated by MATLAB, which plots the Stress Concentration Factor (Kt) against the non-dimensional ratio d/b for both loading scenarios.

The MATLAB script also generated a tabular summary of the results. The data for **Task 1 (Unloaded Hole)** showed a smooth curve, with Kt values ranging from approx 2.76 (at d/b=0.075) down to approx 2.13 (at d/b=0.6).

The digitized theoretical data for **Task 2 (Pin Loaded Hole)** produced a much steeper curve, with Kt values starting as high as 11.0 (at d/b=0.075) and sharply decreasing to 2.40 (at d/b=0.6).



Discussion/Comparison with Literature

The results clearly demonstrate the significant impact of loading conditions on stress concentration.

1. **Task 1 (Unloaded Hole):** ANSYS simulation results for the unloaded hole (Task 1) showed correlation with the theoretical Kt curve found in the literature . The data points followed the expected trend validating the FEA model, meshing strategy, and boundary conditions for the linear static analysis.

2. **Task 2 (Pin-Loaded Hole):** The theoretical Kt values for the pin-loaded case are dramatically higher than for the unloaded case, especially at small d/b ratios. This is because the load is introduced directly at the hole's edge, creating a bearing stress that compounds with the net-section tensile stress. In the unloaded case, the stress "flows" more smoothly around the hole.
3. **Simulation Challenges (Task 2):** While our Task 1 simulation was successful, simulating Task 2 proved highly challenging. The non-linear contact analysis was extremely sensitive to rigid body motion and required careful stabilization (constraining 5 DOFs on the pin) to achieve convergence. Furthermore, initial attempts with the ANSYS Student License failed to produce accurate results for smaller diameters. The Kt values were artificially low, as the coarse mesh could not capture the sharp stress peak.